CORNELL UNIVERSITY
Mechanical and Aerospace Engineering

M&AE 6760: Model Based Estimation

Prof. M. Campbell | **Problem Set #2** | Date: 6 Feb 2026
Spring 2026 | Due: (5:00pm) 27 Feb 2026

Note that suggested MATLAB *.m commands are given in some problems. These are only suggestions, as there may be other commands or Python equivalents you could use.

Problem #1: **Least Squares Parameter Estimation**
Consider a bridge circuit, under constant voltage (measured in mV). Given three measurements, one differential and the other two absolute voltages, we collect the following measurement data for the circuit:

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 10 & 1 \\ 1 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

An m-file is provided for this problem on Canvas. Please complete the following:

a) Given just the measurement data (20 data points), calculate the least squares estimate of $\hat{x}_1, \hat{x}_2$ using only the measurement data. This case assumes that we do not know the sensor error covariance.

b) Assuming a model of the sensor noise v_1, v_2, v_3 as uncorrelated, zero mean, and Gaussian noise with the following statistics for the kth sample:

$$\mathbf{v}_k \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 10^2 & 0 & 0 \\ 0 & 1^2 & 0 \\ 0 & 0 & 100^2 \end{bmatrix} \right) = \mathcal{N}(\mathbf{0}, R_{vb})$$

calculate the (noise) weighted least squares estimate of $\hat{x}_1, \hat{x}_2$ (note that this will technically be called the 'sample covariance'). Compare with your estimate results in (b) with those from (a) and the true value. Discuss what types of errors may arise in either/both cases.

c) Now, you are to analyze the accuracy of your estimates. Find the error covariance of the estimate from (b), defined as $P_{\hat{x}_b}$. Discuss how the errors from (b) relate to this covariance. For the case in (a), can you calculate a similar error covariance? Can you say anything about the error in a formal way (i.e. probabilistically)?

d) Continuing with (a), we can use the results from (a) to find an 'estimate' of the sensor noise covariance, and then improve our parameter estimate. More specifically, using the data $\mathbf{z}_k$ and the estimate from (a) $\hat{\mathbf{x}}_a$, calculate an estimate for the noise covariance $\hat{R}_{va} = \mathsf{cov}(\mathbf{v}_a)$. Using this estimated noise covariance, re-calculate the weighted least squares estimate (i.e. repeat (b) using $\hat{R}_{va}$), and compare and discuss your results with those from (a).

e) Assume now that the three noisy measurements are zero mean and uncorrelated; however, z_1, z_2 are Gaussian and z_3 is uniform with the following statistics:

$$\begin{aligned} (v_1)_k &\sim \mathcal{N}(0, 10^2) \\ (v_2)_k &\sim \mathcal{N}(0, 1^2) \\ (v_3)_k &\sim \mathcal{U}(-173, +173) \end{aligned}$$

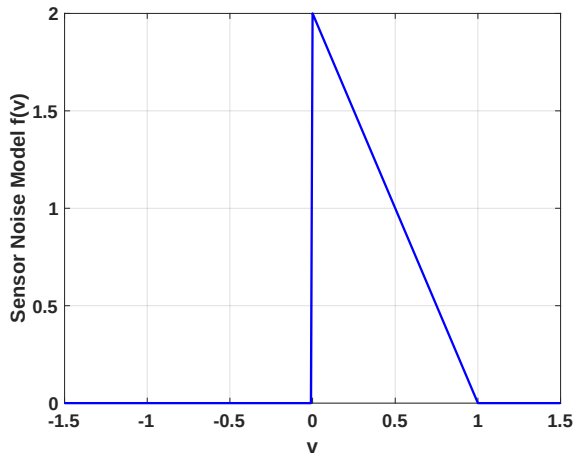
Thus, the problem is the same as (b) except z_3 has uniform sensor noise. Which form of the least squares estimator should you use? Justify your answer. How will the results compare to those from (a) and/or (b)? (note that you do not have to solve, but you can to support your hypothesis).

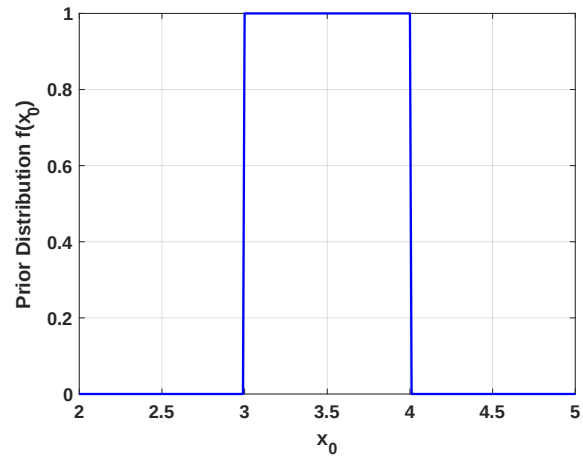
f) If only one measurement z_1, z_2 or z_3 was available, can we find estimates for $\hat{x}_1, \hat{x}_2$? Why or why not?

Problem #2: **Single Parameter Estimation Analytically or via MATLAB functions**
Consider the estimation of a constant via the measurement

$$z = x + v$$

where x is a constant (true value of 3.25), and v is white noise with a pdf given in the figure below (left). Note that this pdf function is $f(v) = [2 - 2v]$, $v \in [0, 1]$. A prior for some of the problems is given below (right).





An m-file is given on Canvas for this problem. Note that you can solve these problems analytically, or using MATLAB functions as shown in the file (e.g., makedist.m , pdf.m , etc.) But not via simulation of samples. Please answer the following:

a) Given a single measurement $z = 3.25$, find the weighted least squares (LS) and maximum likelihood (ML) estimates. How do they compare? *Some comments*: The LS solution we derived in class assumed zero mean sensor noise, and the above pdf for the noise is not zero mean. The likelihood function with Gaussians was always symmetric, so take care to think through signs in your solution.

b) Given the same measurement, and now a prior on x (above right), find the maximum a posteriori (MAP) and minimum mean square (MMSE) estimates. How do they compare? What is the impact of the prior?

c) Discuss how the LS and MMSE variances change as measurements are taken. How many measurements are required such that the variance of the MMSE estimate is within 1% of the variance from the LS estimate?

Problem #3: Discrete Classification of Objects in a Scene

Consider a deep learning detector which classifies objects in a scene from a self-driving car. The following are discrete state values x^i for objects, as well as measurement values z^j:

x^1:	car object	z^1:	car object
x^2:	pedestrian object	z^2:	pedestrian object
x^3:	cyclist object	z^3:	cyclist object

The sensor likelihood model $p(z|x)$ can then be given as a table of options $p(z^j|x^i)$:

	z^1	z^2	z^3
x^1	0.8	0.1	0.1
x^2	0.1	0.6	0.3
x^3	0.1	0.3	0.6

Time k is denoted as subscript $()_k$, i.e. $z_{10} = z^1$ is the measurement of a 'car object' at $k = 10$. Please answer the following:

a) In your own words, describe the discrete likelihood function including how the variables map, what the relative values mean, and a simple example (no need for more than one paragraph)

b) If the first measurement is of a pedestrian, i.e., $z_1 = z^2$, calculate the likelihood function and the ML estimate.

c) Given the same measurement, if the prior is uniform, i.e., $p(x_0) = [0.33, 0.33, 0.33]$, calculate the MAP estimate. How does your estimate compare to the solution from (b)?

d) Repeat (c), but for a prior related the approximate ratio of objects you see in a day, which is $p(x_0) = [0.85, 0.1, 0.05]$. Compare (c) and (d) and discuss the impact of the prior on the solution.

e) Consider now the case of two measurements $z_1 = z^2$, $z_2 = z^2$, calculate the ML estimate. Compare (b) and (e) and discuss the impact of the second measurement.

f) Estimate (using MATLAB/Python) how many measurements you would need to be 99% sure that the object was a pedestrian (via ML estimation). Hint: you can normalize the likelihood to make it a true pdf distribution.

Problem #4: MMSE Estimate of a function

Consider discretely measured process of the following form:

$$z_k = x_1 + x_2 \sin(10t_k) + x_3 e^{2t_k^2} + v_k, \quad k = 1, 2, \cdots, 10$$

with t_k starting from 0 and sampled every 0.1 seconds. The true values (x_1, x_2, x_3) are (1, 1, 1) and the measurement errors v_k are zero mean, Gaussian random variables with a covariance R_0 of diagonal standard deviations:

$$\begin{array}{lllll} \sigma_1 = 0.001 & \sigma_2 = 0.002 & \sigma_3 = 0.005 & \sigma_4 = 0.010 & \sigma_5 = 0.008 \\ \sigma_6 = 0.002 & \sigma_7 = 0.010 & \sigma_8 = 0.007 & \sigma_9 = 0.020 & \sigma_{10} = 0.006 \end{array}$$

You will find an m-file on the Canvas website simulating this process to create the measurements, as well as a plotting script that may be helpful.

You are given an *a priori* parameter estimate and covariance

$$\mathbf{x}_0 = \begin{bmatrix} 1.01 \\ 0.98 \\ 0.99 \end{bmatrix}, \quad P_0 = \begin{bmatrix} 0.001 & 0 & 0 \\ 0 & 0.001 & 0 \\ 0 & 0 & 0.001 \end{bmatrix}$$

Please complete the following:

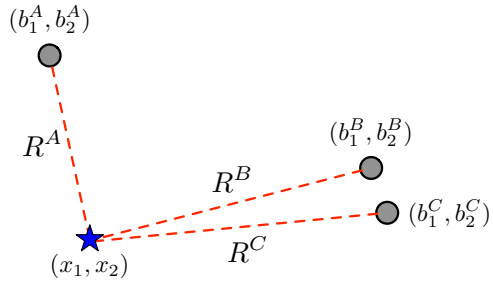
a) Solve for the MMSE estimate and error covariance for $\mathbf{x} = (x_1, x_2, x_3)^T$. Note that this formulation is slightly different from that presented in class in that H_k varies.

b) Perform a parametric study with different sensor noise, defined as $R_{\text{noise}} = \alpha R_0$, where R_0 is the nominal case, and $\alpha \in (10^{-3}, 10^{-2}, 10^{-1}, 10^0, 10^1, 10^2, 10^3)$. Discuss the impact of a different noise.

c) Repeat the parametric study with constant sensor noise, but varying prior covariance $P_{\text{prior}} = \alpha P_0$. Discuss the impact of a different prior.

Problem #5: Least Squares for a Nonlinear Ranging Problem

Consider the problem of localizing yourself via your cell phone using range measurements to beacons, such as cell phone towers. A 'top down' view of this problem is given below:



where the true location of the car is $(x_1 = -5, x_2 = 2)$, and the true location of the beacons is $b_1^{(\cdot)}, b_2^{(\cdot)}$.

Up to three range measurements are collected at any one time (subset of A,B,C). The three range measurements are assumed to be corrupted with addititive, Gaussian white noise, i.e.

$$\mathbf{z}_k = \begin{bmatrix} R^A \\ R^B \\ R^C \end{bmatrix} + \mathbf{v}_k$$

An m-file is given for this problem on Canvas. Please answer the following:

a) For a noise covariance of

$$R_{\mathbf{v}_k} = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 10 \end{bmatrix}$$

calculate the 2D location estimate $(\hat{x}_1, \hat{x}_2)$ and error covariance via nonlinear least squares after 10 measurements of all three beacons are made. You can initialize your estimator at (0,0);

b) Now, consider two cases when 10 measurements to only two beacons are available: (A,B) and (B,C). Estimate and compare the location and error covariance with each other, and with part (a). Why do they differ in performance?

c) Let's evaluate the impact of the linearization. Calculate the Jacobian by linearizing around the 'true' location $(x_1 = -5, x_2 = 2)$ (i.e. a perfect linearization, which yields a constant H matrix). Repeat the two cases from (b) using the initial condition (0,0) and compare the numerical results to (b). Does the linearization impact the results? Note that you should be able to find an analytical solution to the error covariance in these cases (c) given that we have a constant H matrix.

d) Repeat (a), but for the case when the noise covariance is correlated (perhaps from a local power supply near B,C)

$$R_{\mathbf{v}_k} = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 10 & 9 \\ 0 & 9 & 10 \end{bmatrix}$$

Discuss any changes/trends that result from this covariance (compared to (a)).